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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHE 513 — Past Questions & Materials
by Renance

Course: CHE 513
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 500

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Paper 1 of 2 — CHE 513 - 2023/2024 - Second Semester - Chemical Process Technology

Level 500 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF CHEMICAL ENGINEERING

First Semester Examination, 2023/2024 Academic Session

COURSE TITLE: CHEMICAL PROCESS TECHNOLOGY,

COURSE CODE: CHE 513

CREDIT UNITS: 3

INSTRUCTIONS: Answer Any Five Questions

TIME ALLOWED: 3 Hours

Start each Question on a New Page.

1a (i) As process engineer explain concisely what you understand by filter backwashing? When to backwash?

(ii) Draw and label a flow chart for a demineralized plant

(iii) Give 5 examples of Electrodeionization (EDI)

1b (i) In the design parameter of a clarifier, what are the two most considered parameters?

(ii) Write an equation how chlorine is used in treatment of water

(iii) List 5 merits and 3 demerits of Electrodeionization (EDI)

2a (i) Draw a well label diagram of how water is filtered. Stating the 4 mechanism of solids removal.

(iii) Explain briefly the meaning of nitrification and denitrification. Differentiate between them and mention the microorganisms involved in each.

2b (i) What is the economy of phosphate remove from waste water using chemical treatment. Write the equation for the reaction.

(ii) Draw a block diagram of a complete waste water treatment plant.

3a (i) How do treated water get contaminated with heavy metals? As a chemical engineer suggest any possible way of handling this issue.

(ii) Sewer fee is typically ₦30/m³, an additional ₦0.60/kg BOD and ₦2.00/kg SS are charged for discharges where the BOD is more than 350 mg/L and the suspended solids (SS) are more than 400mg/L. A poultry processing facilities utilizes 3000m³ of water per day and releases effluent with SS = 1630 mg/L and BOD = 570 mg/L. What is the cost of disposing of wastewater every day at the plant.

(iii) As a process engineer succinctly describes the two techniques for demineralizing water.

3b (i) How do environmental and process conditions handle the accumulation of sludge in a clarifier/settler design.

(ii) Explain concisely the 3 ways in which ion-exchange technology can be used in water treatment.

(iii) Explain briefly what you understand by EDI

- 4a (i) Explain concisely using Lead-Chamber process, how sulphuric acid is manufactured
(ii) write an equation for reaction in (4ai)
(iii) Draw a block diagram of how glass is manufactured
- b (i) List 10 classes of trademarks
(ii) using contact process. Explain and write all the equations of the reaction how SO_2 is prepared
(iii) Explain the type of chemical patent you know
- 5a (i) As an engineer that want to assign his trade make to another person or entity for usage within a period of time what are the basic requirements for trademarks recording
(ii) Draw a flow chart of how to prepare sulphuric acid.
(iii) what do you understand by chemical patent
- b (i) Explain briefly the process of manufacturing of sulphur. Draw the diagram for the process.
(ii) As an intending applicant, what are the basic requirements for the registration of trademarks
(iii) Explain what you understand by research and development
- 6a (i) Explain concisely the process of desalination of water
(ii) write an equation on how plant make sugar
(iii) Draw a block diagram of refining sugar
(iv) Write the main reaction in the production of urea
- b (i) Write 5 difference between wet and dry process in cement production
(ii) What are the various steps involved in cement production
(iii) Draw a flow chart of urea production
- 7a (i) Draw a flow chart for demineralized plant
(ii) Draw a block diagram for ammonia production and the raw materials for the processes
(iii) Write an equation for the reaction
- b (i) Explain briefly how water gas is manufactured. Write all the equations for the reaction
(ii) Write 5 characteristics of a good varnish
(iii) What are the 3 substances needed to enable the extraction of iron from its ore.
(iv) write the 5 equations of the reaction that takes place before iron is produced

Paper 2 of 2 — CHE 513 CHEMICAL PROCESS TECHNOLOGY

Level 500 • First Semester • PastQuestion

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